A Feasibility-Aware HUBO/QUBO Benchmark for UAV Obstacle-Avoidance with Visibility Cost

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Abstract

Quantum-assisted UAV path planning has recently been studied using QAOA-style obstacle-avoidance formulations, but a low-energy binary sample is not necessarily a valid UAV trajectory. This paper presents a feasibility-aware benchmark for a discrete 8×8, $L = 20$ UAV grid-planning problem with obstacle avoidance, a line-of-sight visibility cost, and a higher-order temporal buffer-risk term. The horizon L denotes discrete planning layers, not physical seconds. The model uses binary variables $x_{t,v}$ indicating whether the UAV occupies cell v at time layer t and forms a HUBO objective with one-hot, start, motion, obstacle, visibility, terminal-goal, and sustained near-obstacle proximity terms. The instance has 1,280 original binary variables and 118,344 polynomial terms, including 4,392 cubic monomials. We compare A*, RRT-best, a QUAV-style QAOA candidate-path selector, native trajectory-level simulated annealing (SA), the classical `neal` BQM sampler after HUBO-to-QUBO reduction, and an exact CP-SAT baseline that minimizes the native HUBO soft objective over hard-feasible paths. The CP-SAT solver returns OPTIMAL and certifies that the best observed SA path is globally optimal for the enforced feasible-path objective, with implementation HUBO energy -20752.43, soft cost 247.57, path length 14, six turns, and 85% visibility. On the tested instance, the results show why feasibility-aware decoding, exact baselines, and quadratization effects matter; they do not establish quantum speedup or performance across an instance suite.

Code — https://github.com/alimohammadi44/uav-hubo-qiml-2026

Introduction

UAV path planning in cluttered environments requires at least two safety properties: the path must avoid obstacles, and the decoded plan must be executable under the allowed motion model. When perception or tracking is relevant, the planner may also prefer cells with line of sight to the target. Recent quantum-assisted work such as QUAV (Innan et al. 2025) formulates UAV obstacle-avoidance path planning using QAOA-style optimization and compares against A* and RRT. QUAV is therefore an important closest reference for quantum-assisted UAV navigation. However, a QAOA, annealing, or QUBO solver returns a binary sample, and the sample must be decoded and checked before it can be called a valid path.

This paper uses a deliberately compact grid so that modeling and solver behavior can be inspected directly. Rather than claiming quantum speedup, it defines a reproducible HUBO/QUBO-compatible benchmark on one fully specified instance, separates energy minimization from path validity, and compares classical, quantum-inspired, and quantum-compatible solver pathways.

Problem Statement and Scope

The planning problem considered here is a static, discrete visibility-aware path-planning abstraction. A UAV must move from a fixed start cell to a fixed target/goal cell through legal grid transitions while avoiding obstacle cells. Visibility is represented as a soft line-of-sight cost from each candidate UAV cell to the fixed target cell. Thus, this paper does not solve continuous quadrotor control, full moving-target tracking, or camera field-of-view control. The abstraction is intentionally small enough that the encoded binary model, decoded paths, feasibility failures, and exact classical baselines can be inspected in detail. The time horizon L is a design parameter giving the number of discrete planning layers. It is not assumed to be seconds unless a physical sampling time Δt is separately assigned.

The contributions are as follows. First, we give a complete binary decision-variable formulation for UAV grid planning with obstacle avoidance and a visibility cost. Second, we identify the real source of higher-order structure: a temporal buffer-risk term, not the static line-of-sight cost. Third, we introduce a feasibility-aware benchmark that reports structural feasibility, goal arrival, and full-task success separately. Fourth, we implement a same-platform QUAV-style QAOA candidate-path baseline, clearly not as official QUAV code but as a reproducible reference following the published workflow. Fifth, we add an exact CP-SAT implementation, enabling global comparison against the native HUBO soft objective over hard-feasible paths.

The novelty is narrowly the feasibility-aware evaluation layer around a quantum-compatible UAV HUBO/QUBO formulation. The same decoded paths are compared under graph-search, candidate-path QAOA, native trajectory-SA,

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quadratized BQM sampling, and exact CP-SAT, with validity and goal arrival separated from energy. The paper claims neither a faster planner than A* or CP-SAT nor quantum advantage; without exact baselines and decoded-path checks, low-energy samples can be misleading.

Related Work

Classical and Visibility-Aware UAV Planning

Classical UAV planning spans graph search, sampling-based planning, trajectory optimization, and perception-aware control. A* remains a standard shortest-path reference on discretized maps (Hart, Nilsson, and Raphael 1968), while RRT provides a widely used sampling-based alternative (LaValle 1998). For target-tracking and perception-aware flight, Chen *et al.* formulate moving-target tracking in cluttered environments (Chen, Liu, and Shen 2016); Hausman *et al.* study occlusion-aware multi-robot 3D tracking (Hausman et al. 2016); Penin *et al.* combine aggressive target tracking with collision and occlusion avoidance (Penin, Robuffo Giordano, and Chaumette 2018); and Falanga *et al.* incorporate perception objectives into model-predictive control (Falanga et al. 2018). Wang *et al.* extend visibility-constrained tracking to multiple UAVs (Wang et al. 2022), while Masnavi *et al.* derive a real-time multi-convex MPC for occlusion-free target tracking with collision avoidance (Masnavi et al. 2022). These continuous-control works motivate the visibility term used here, but our problem is deliberately a static discrete evaluation instance rather than a moving-target control formulation.

Quantum and QUBO Formulations for UAV/Drone Optimization

Quantum optimization for routing has developed rapidly; QUBO provides a standard binary optimization representation (Glover, Kochenberger, and Du 2019), while QAOA is a widely used variational gate-based approach (Farhi, Goldstone, and Gutmann 2014). A systematic review by Osaba *et al.* surveys early quantum-routing formulations and highlights recurring issues of encoding size, constraints, and fair classical comparison (Osaba, Villar-Rodriguez, and Oregi 2022). Several UAV-specific studies are especially relevant. Huang *et al.* use variational quantum algorithms, including QAOA and VQE, to select collision-avoiding routes for multiple UAVs from predefined alternatives (Huang et al. 2022). Davies and Kalidindi formulate constrained multi-UAV mission planning as MILP and QUBO and study quantum annealing, QAOA, and VQE routes (Davies and Kalidindi 2024). Hua *et al.* investigate QUBO-based drone route planning with quantum/digital annealers and compare against Gurobi on wind-influenced urban routing instances (Hua et al. 2024; Tarquini et al. 2026).

More recent UAV-specific frameworks move closer to the present setting. Q4DR combines a QAOA clustering stage with annealing-based drone routing under asymmetric costs, forbidden arcs, and charging constraints (Osaba et al. 2025). QUAV uses QAOA-style optimization for single-UAV obstacle-avoidance path planning and reports both simulation and quantum-hardware experiments; it is the closest prior work to our candidate-path QAOA baseline (Innan et al. 2025). Qiu *et al.* formulate multi-target UAV path planning as QUBO and solve it using a coherent Ising machine, providing another direct example of binary optimization for UAV navigation (Qiu et al. 2025). Barbeau *et al.* formulate 1D and 3D target tracking, including obstacle-distance constraints, as constrained quadratic models for quantum annealing (Barbeau, Janabi-Sharifi, and Masnavi 2025). These studies demonstrate multiple quantum-compatible routes for UAV routing, collision avoidance, and tracking, but they optimize different abstractions and generally do not center evaluation on decoded structural feasibility under a common native objective.

QUBO Path Finding, Multi-Agent Planning, and the Gap Addressed Here

Beyond UAV-specific work, recent multi-agent path-finding studies are relevant to encoding and scalability. Gerlach *et al.* propose an optimal hybrid quantum-classical MAPF method based on branch-and-cut-and-prize with iteratively solved QUBO conflict subproblems (Gerlach et al. 2025). Gonzalez Villasmil proposes a robotics-oriented time-indexed QUBO with BFS-based logical preprocessing, adaptive penalties, and time-window decomposition for multi-robot grid planning (González Villasmil 2026). These works reinforce two lessons that directly motivate our benchmark: path-planning encodings can grow rapidly with space-time discretization, and solver outputs must be interpreted relative to hard feasibility and strong classical references.

Among the works reviewed above, we did not find a study combining a single-UAV time-indexed HUBO with a static line-of-sight cost, a genuinely cubic sustained near-obstacle term, HUBO-to-QUBO conversion, decoded feasibility/goal/full-success reporting, and exact CP-SAT on the same native objective.

Discrete Planning Model

Let $G = (V, E)$ be an $N \times N$ grid graph. In the experiments $N = 8$, $|V| = 64$, and time layers are indexed $t = 0, \ldots, L-1$ with $L = 20$. Each time layer is one discrete trajectory frame; a larger L gives the planner more room for detours or hovering but increases the number of variables linearly. Let $O \subset V$ be the obstacle set, $s \in V$ the start cell, and $g \in V$ the target cell. The permitted next-cell set is

$$\mathcal{N}(u) = \{u\} \cup \{v \in V : (u, v) \in E\}, \tag{1}$$

where the self-loop represents hovering.

The binary state variable is

$$\begin{gathered} x_{t,v} \in \{0, 1\}, \\ x_{t,v} = 1 \Leftrightarrow \text{the UAV is at cell } v \text{ at time } t. \end{gathered} \tag{2}$$

A valid binary solution decodes to a trajectory $\pi = (v_0, \ldots, v_{L-1})$ only if exactly one state variable is active at every layer.

Let $\mathrm{LOS}(u, g)$ denote the rasterized Bresenham line cells strictly between u and g. The per-cell visibility/occlusion-depth surrogate is

$$C_{\mathrm{occ}}(u) = |\mathrm{LOS}(u, g) \cap O|. \tag{3}$$

A zero value indicates clear line of sight in this grid model. Let $B \subset V \backslash O$ be the set of non-obstacle cells within Chebyshev distance one of an obstacle. Cells in B are legal but risky.

HUBO Formulation

The total polynomial objective is

$$\begin{aligned} H(x) =& H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} \\ &+ H_{\text{occ}} + H_{\text{prox}} + H_{\text{goal}} + H_{\text{term}}. \end{aligned} \tag{4}$$

The structural terms are

$$H_{\text{uniq}} = \lambda_u \sum_{t=0}^{L-1} \left(\sum_{v \in V} x_{t,v} - 1 \right)^2, \tag{5}$$

$$H_{\text{start}} = \lambda_s (1 - x_{0,s}), \tag{6}$$

$$H_{\text{move}} = \lambda_m \sum_{t=0}^{L-2} \sum_{u \in V} \sum_{v \notin \mathcal{N}(u)} x_{t,u} x_{t+1,v}, \tag{7}$$

$$H_{\text{obs}} = \lambda_o \sum_{t=0}^{L-1} \sum_{v \in O} x_{t,v}. \tag{8}$$

The static visibility cost is linear:

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_{t=0}^{L-1} \sum_{u \in V} C_{\text{occ}}(u) x_{t,u}. \tag{9}$$

The sole cubic term is the sustained-proximity penalty:

$$\begin{aligned} H_{\text{prox}} = \lambda_p \eta \sum_{t=1}^{L-2} \sum_{u \in B} \sum_{v \in B \cap \mathcal{N}(u)} \\ \sum_{w \in B \cap \mathcal{N}(v)} x_{t-1,u} x_{t,v} x_{t+1,w}. \end{aligned} \tag{10}$$

Here $\eta = 3$. This term penalizes remaining in connected buffer cells for three consecutive layers. Thus the model is genuinely HUBO, but the higher-order degree comes from temporal buffer-risk modeling rather than from static visibility.

The goal terms are

$$H_{\text{goal}} = \lambda_g \sum_{t=0}^{L-1} \sum_{v \in V} \frac{t+1}{L} \|p_v - p_g\|_2 x_{t,v}, \tag{11}$$

$$H_{\text{term}} = \lambda_{\text{term}} \sum_{v \neq g} x_{L-1,v}. \tag{12}$$

The updated instance uses $\lambda_u = \lambda_s = \lambda_m = \lambda_o = 1000$, $\lambda_{\text{occ}} = 10$, $\lambda_p = 5$, $\lambda_g = 8$, $\lambda_{\text{term}} = 50$, and $\eta = 3$.

Feasibility-Aware Evaluation

A solver output is not scored only by total HUBO energy. Each binary sample is decoded and checked directly. We report: structural feasibility, meaning one-hot occupancy, correct start, legal transitions, and no obstacle occupancy; goal arrival, meaning $v_{L-1} = g$; and full-task success, meaning both conditions hold for the same decoded sample. Soft cost and implementation HUBO energy are then interpreted on valid decoded paths.

This distinction is essential for QUBO/BQM and QAOA-style workflows. A sample can have a low reduced-model energy while failing one-hot, motion, obstacle, or target-arrival checks. Such a sample is not a UAV plan. The benchmark therefore reports feasibility rates and representative decoded trajectories, not energy alone.

Quantity	Value
Grid size	8×8
Horizon L	20
Original binary variables	1,280
Total polynomial terms	118,344
Linear terms	1,280
Quadratic terms	112,672
Cubic terms	4,392
Obstacles	6
Buffer cells	24

Table 1: Tested HUBO instance.

Solvers and Implementation

All methods were executed on the same Apple M2 Max platform with 32 GB unified memory. Fixed stochastic seeds are recorded in the public implementation; the rates are descriptive for this single instance.

A* and RRT Baselines

A* searches directly on the grid with obstacle avoidance and a heuristic path cost. RRT-best is a randomized grid planner used as a sampling-style reference. These methods return explicit paths and are checked with the same feasibility evaluator. The RRT-best runtime includes 40 fixed-seed trials and best-cost selection.

QUAV-Style QAOA Baseline

QUAV maps UAV path segments to qubits and applies QAOA-style optimization with obstacle-buffer costs. Since no official public code was available in this workflow, we implemented a QUAV-style candidate-path selector following the published algorithmic description. It generates collision-free candidate paths, assigns a QUAV-style geometric cost using length, buffer steps, and turns, and uses a small QAOA-style one-hot selector over the candidates. This baseline is a same-platform reference only, not an exact reproduction of the authors' implementation.

Trajectory-Level SA and `neal`

Trajectory-level SA searches over cell sequences and evaluates the implementation HUBO energy directly, including the cubic term. Its proposal operator preserves one-hot occupancy, start, legal motion, and obstacle avoidance by construction. In contrast, `neal` is a classical BQM simulated-annealing sampler (D-Wave Systems Inc. 2026b). Therefore the HUBO is first reduced to a QUBO/BQM using

Method	Feas.	Goal	Full	Len.	Turns	Buf.	Vis.	HUBO E	Time/run
A*	Yes	Yes	Yes	14	7	4	85.0%	-20742.26	0.028s
RRT	Yes	Yes	Yes	14	5	4	70.0%	-20700.19	0.0011s
QUAV-QAOA	Yes	Yes	Yes	14	1	3	80.0%	-20725.94	4.11s
CP-SAT exact	Yes	Yes	Yes	14	6	4	85.0%	-20752.43	1.02s
HUBO-SA	100.0%	16.0%	16.0%	14	6	4	85.0%	-20752.43	24.15s
QUBO/`neal`	12.0%	8.0%	0.0%	18	12	2	20.0%	-20060.47	0.123s

Table 2: Unified same-platform comparison. Yes/no is a single output; percentages use 50 runs/reads. SA reports its best full-success path; `neal` reports its best-energy feasible non-goal sample. Time/run is mean wall-clock time; the `neal` value is amortized over one 50-read batch. Only `neal` samples the full penalized QUBO; the other methods pre-enforce feasibility, so solver difficulty is not like-for-like.

`dimod` (D-Wave Systems Inc. 2026a); cubic monomials require auxiliary-variable quadratization, following the standard reduction principle of Rosenberg (Rosenberg 1975), before sampling.

Exact CP-SAT Baseline

We implemented an exact CP-SAT model using OR-Tools (Perron and Furnon 2026). The model enforces exactly one cell per layer, the fixed start, final target arrival, obstacle avoidance, and legal transitions. Auxiliary binary variables linearize each cubic proximity monomial $y_{tuvw} = x_{t-1,u}x_{t,v}x_{t+1,w}$. The objective minimizes

$$H_{\text{occ}} + H_{\text{prox}} + H_{\text{goal}} + H_{\text{term}} \tag{13}$$

over the hard-feasible set. In the solver implementation, the additive constant $\lambda_u L + \lambda_s = 21000$ is omitted. Consequently, the reported implementation energy is $\widetilde{H} = H - 21000$. For a hard-feasible path with soft cost 247.57, this gives $\widetilde{H} = -20752.43$. CP-SAT returned OPTIMAL; therefore, its solution provides a globally optimal reference for the implemented hard-feasible objective.

Results

The quantitative results are organized in two complementary tables. Table 1 summarizes the tested HUBO instance, including its grid, horizon, polynomial size, and obstacle geometry. Table 2 compares all six methods on that instance using feasibility, path quality, implementation HUBO energy, and mean runtime.

In Table 2, **Feas.** means one-hot occupancy, correct start, legal transitions, and obstacle avoidance; **Goal** means final-layer target arrival; and **Full** requires both. **Len.** counts non-hover moves, **Turns** direction changes excluding hovering, **Buf.** trajectory layers in obstacle-buffer cells, and **Vis.** layers with clear target line of sight. **HUBO E** is implementation energy with the constant 21,000 omitted.

Decoded Optimized Paths

Fig. 1 shows the actual decoded paths returned by each method on the same map. This figure is important because the numerical table alone does not reveal how the methods use the obstacle-buffer region or why different objectives prefer different routes. A*, RRT-best, QUAV-style QAOA, CP-SAT, and the best full-success SA trajectory all reach the target without collisions. The `neal` panel shows the best-energy decoded QUBO sample; it is structurally feasible but fails to reach the target, explaining why its full-task success rate is zero.

The exact CP-SAT baseline reports OPTIMAL, with soft cost 247.57 and implementation HUBO energy -20752.43. This matches the best trajectory-SA energy, showing that SA found a globally optimal path for the hard-feasible exact objective on this instance, but SA reached the target in only 8 of 50 runs. A* is fully successful, but its implementation HUBO energy is slightly worse because it optimizes a graph-search surrogate rather than the exact cubic proximity-aware objective.

The QUAV-style QAOA selector produces a valid shortest path with only one turn and the lowest QUAV-style geometric cost among its candidates. However, under the native HUBO objective its energy is -20725.94, worse than the CP-SAT/SA optimum. This illustrates a central point: a QUAV-style geometric objective and the native HUBO objective are not the same, even when both return collision-free paths. `neal` after quadratization remains the weakest method in this run: it has 6/50 structural feasibility and 4/50 goal arrival, but 0/50 full-task success.

Penalty Sweep

The penalty-weight sweep for `neal` shows that weak structural penalties permit low-energy but infeasible samples. In the updated run, structural feasibility was $0/15$ for $\lambda \in \{50, 100, 300, 600\}$, $4/15$ at $\lambda = 1000$, and $5/15$ at $\lambda = 2000$. The best feasible soft cost improved from 915.4 to 590.6 between $\lambda = 1000$ and $\lambda = 2000$, still worse than A*, SA, and CP-SAT. Because target arrival depends on penalty balance, the observed zero full-task success cannot be attributed to the sampler alone.

Discussion

The results show why the paper's contribution is not a simple claim that quantum computing outperforms classical planning. On this instance, exact CP-SAT proves the optimum of the hard-feasible native-HUBO soft objective, A* is fully successful, and trajectory-SA can find the exact optimum but with poor target-arrival reliability. The decoded paths in Fig. 1 make the solver differences explicit. The QUAV-style QAOA baseline is useful because it shows that a quantum-

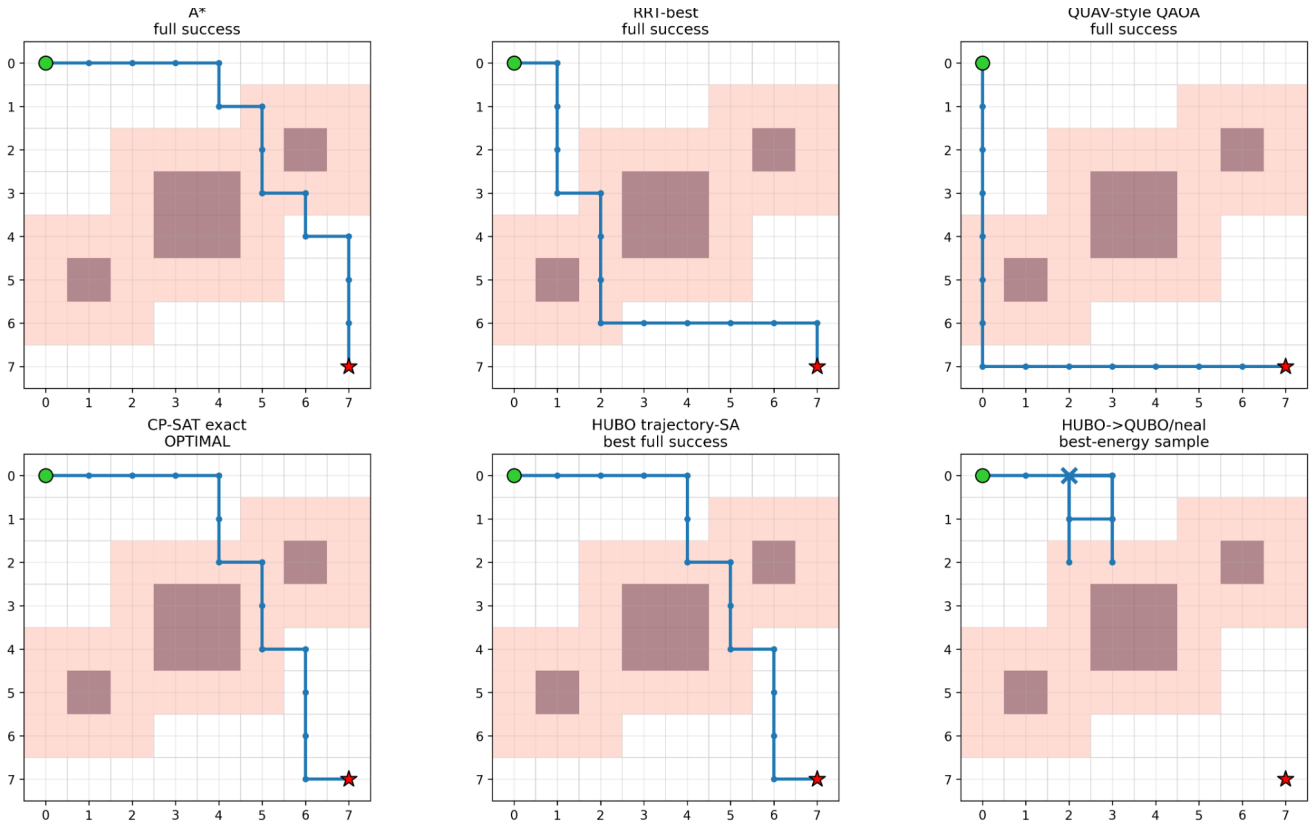


Figure 1: Decoded paths on the common 8×8, $L = 20$ evaluation instance. Dark cells are obstacles, light cells are obstacle-buffer cells, the green circle is the start, and the red star is the target. CP-SAT is globally optimal for the hard-feasible native-HUBO soft objective. The structurally feasible `neal` sample does not reach the target, illustrating a quadratization/sampling failure mode.

assisted geometric selector can produce a valid obstacle-free path on the same platform. However, the native HUBO objective distinguishes visibility, temporal buffer-risk, and terminal behavior in a different way. Therefore, same-platform decoded-path evaluation is necessary for fair comparison.

The exact CP-SAT formulation also clarifies the role of quantum-compatible solvers. If a QUBO sampler is used, it must overcome not only path planning but also one-hot, movement, obstacle, target, and quadratization constraints. The benchmark exposes these failure modes by reporting structural feasibility, goal arrival, and full-task success separately.

The QUAV-style row in Table 2 should be interpreted carefully: it is a candidate-path selector over generated paths, not a native solution of the full $x_{t,v}$ HUBO. This candidate-path route is practical because feasibility is mostly handled before optimization, but it optimizes over a restricted candidate set. A* and CP-SAT remain correctness anchors. If a quadratic sampler is used, the cubic term must first be reduced to quadratic form with auxiliary variables and additional penalties. This reduction is included as a conceptual conversion pathway and a tested failure mode rather than as a complete quadratization-accounting claim.

Limitations and Future Work

This single static grid omits dynamics, delays, field-of-view constraints, and an official QUAV implementation, so its rates are not distribution-level estimates. Here, "benchmark" denotes the reproducible single-instance evaluation used in this study, not an instance suite. Future work will develop a broader suite varying grid size, obstacle density and placement, horizon, start/goal locations, and random seeds, with sensitivity and statistical analyses. It will also vary terminal, structural, and quadratization penalties to distinguish sampler from penalty effects; compare cubic and quadratic buffer models; and study matched QUAV, moving targets, and stronger candidate generation.

Conclusion

This paper demonstrates a feasibility-aware HUBO/QUBO benchmark for UAV obstacle-avoidance with visibility cost on one fully specified instance. A cubic term models sustained temporal proximity to obstacle-buffer cells. Comparing A*, RRT-best, QUAV-style QAOA, trajectory-SA, `neal`, and exact CP-SAT shows that geometric path quality, HUBO energy, and trajectory validity are different metrics. The CP-SAT optimum matches the best observed SA

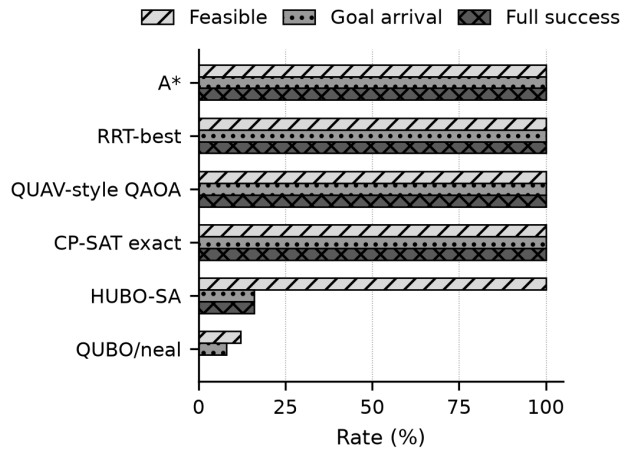


Figure 2: Feasibility-aware success metrics. CP-SAT certifies a full-success optimum; `neal` has nonzero separate feasibility and goal-arrival rates but no full-task success.

path, while quadratized `neal` has zero full-task success in the tested reads.

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References

Barbeau, M.; Janabi-Sharifi, F.; and Masnavi, H. 2025. A Quantum Annealing Approach to Target Tracking. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 3022–3027.

Chen, J.; Liu, T.; and Shen, S. 2016. Tracking a Moving Target in Cluttered Environments Using a Quadrotor. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 446–453.

D-Wave Systems Inc. 2026a. dimod: Binary Quadratic Model Library. Software package.

D-Wave Systems Inc. 2026b. neal: Simulated Annealing Sampler for Binary Quadratic Models. Software package.

Davies, E. G.; and Kalidindi, P. V. 2024. Quantum Algorithms for Drone Mission Planning. In *Proceedings of SPIE Quantum Technologies for Defence and Security*, volume 13202, 1320209.

Falanga, D.; Foehn, P.; Lu, P.; and Scaramuzza, D. 2018. PAMPC: Perception-Aware Model Predictive Control for Quadrotors. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 1–8.

Farhi, E.; Goldstone, J.; and Gutmann, S. 2014. A Quantum Approximate Optimization Algorithm. *arXiv preprint arXiv:1411.4028*.

Gerlach, T.; Lee, L. K.; Barbaresco, F.; and Piatkowski, N. 2025. Hybrid Quantum-Classical Multi-Agent Pathfinding. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 19161–19171.

Glover, F.; Kochenberger, G.; and Du, Y. 2019. Quantum Bridge Analytics I: A Tutorial on Formulating and Using QUBO Models. *4OR*, 17(4): 335–371.

González Villasmil, J. 2026. Scalable Multi-Robot Path Planning via Quadratic Unconstrained Binary Optimization. *arXiv preprint arXiv:2602.14799*.

Hart, P. E.; Nilsson, N. J.; and Raphael, B. 1968. A Formal Basis for the Heuristic Determination of Minimum Cost Paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2): 100–107.

Hausman, K.; Kahn, G.; Patil, S.; Müller, J.; Goldberg, K.; Abbeel, P.; and Sukhatme, G. S. 2016. Occlusion-Aware Multi-Robot 3D Tracking. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 1863–1870.

Hua, R.; Di Lorenzo, D.; Chinesta, F.; and Codognet, P. 2024. Quantum Annealing Solutions for Drone Route Planning Problems. In *Proceedings of the IEEE International Conference on Quantum Computing and Engineering (QCE)*, 600–610.

Huang, Z.; Li, Q.; Zhao, J.; and Song, M. 2022. Variational Quantum Algorithm Applied to Collision Avoidance of Unmanned Aerial Vehicles. *Entropy*, 24(11): 1685.

Innan, N.; Kashif, M.; Marchisio, A.; Gan, Y.-S.; Barbaresco, F.; and Shafique, M. 2025. QUAV: Quantum-Assisted Path Planning and Optimization for UAV Navigation with Obstacle Avoidance. In *Proceedings of the IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 208–215.

LaValle, S. M. 1998. Rapidly-Exploring Random Trees: A New Tool for Path Planning. Technical Report 98-11, Department of Computer Science, Iowa State University.

Masnavi, H.; Adajania, V. K.; Kruusamäe, K.; and Singh, A. K. 2022. Real-Time Multi-Convex Model Predictive Control for Occlusion-Free Target Tracking with Quadrotors. *IEEE Access*, 10: 29009–29031.

Osaba, E.; Miranda-Rodriguez, P.; Oikonomakis, A.; Petrič, M.; Ruiz, A.; Bock, S.; and Kourtis, M.-A. 2025. Solving Drone Routing Problems with Quantum Computing: A Hybrid Approach Combining Quantum Annealing and Gate-Based Paradigms. In *Proceedings of the IEEE Congress on Evolutionary Computation (CEC)*, 1–8.

Osaba, E.; Villar-Rodriguez, E.; and Oregi, I. 2022. A Systematic Literature Review of Quantum Computing for Routing Problems. *IEEE Access*, 10: 55805–55817.

Penin, B.; Robuffo Giordano, P.; and Chaumette, F. 2018. Vision-Based Reactive Planning for Aggressive Target Tracking While Avoiding Collisions and Occlusions. *IEEE Robotics and Automation Letters*, 3(4): 3725–3732.

Perron, L.; and Furnon, V. 2026. OR-Tools. Google, software package.

Qiu, G.; Lin, R.; Zhu, J.; and Ren, J. 2025. Quantum-Driven UAV Path Planning Using QUBO Model and Coherent Ising Machines. In *Proceedings of the 8th International Conference on Advanced Algorithms and Control Engineering (ICAACE)*, 388–393.

Rosenberg, I. G. 1975. Reduction of Bivalent Maximization to the Quadratic Case. *Cahiers du Centre d'Etudes de Recherche Opérationnelle*, 17: 71–79.

Tarquini, S.; Dragoni, D.; Vandelli, M.; and Tudisco, F. 2026. Testing Quantum and Simulated Annealers on the Drone Delivery Packing Problem. *Quantum Machine Intelligence*, 8: 14.

Wang, J.; Wu, Y.-X.; Chen, Y.-Q.; and Ju, S. 2022. Multi-UAVs Collaborative Tracking of Moving Target with Maximized Visibility in Urban Environment. *Journal of the Franklin Institute*, 359(11): 5512–5532.